

Singapore

Competition and Consumer Commission

In 2023, the Competition and Consumer Commission of Singapore (‘CCCS’) established a new Data and Digital (‘D2’) division to lead and focus the agency’s efforts on technology development in big data and digital markets. Below is a list of tools recently rolled out or currently under development by D2.

I. Complaint Analytics Tool (CAT)

The CAT is a python-based tool, developed in-house by D2, which applies natural language processing on a sizeable volume of text passages, and uses machine learning to cluster them into intelligible topics, which are then ranked by popularity. The tool can also compare topics across time and trace the evolution of a particular subject over a period.

Since its introduction, the CAT has found several use cases. First, it has been used to analyse complaints received from the public. The results are regularly presented to senior management for a deeper understanding of trending issues in the marketplace. This also helps to generate leads for potential enforcement cases. Second, the CAT has been applied to analyse feedback received in some public consultations to improve productivity in considering such feedback. Third, the tool has also been applied to analyse entries from the CCCS-Economics Society of Singapore (‘ESS’) essay competition to identify frequently-discussed issues.

II. Fake Review Detection Tool (FRDT)

As CCCS is the enforcement agency for both competition and consumer protection laws, D2 has developed the FRDT to augment our latter function. The FRDT is a python-based machine learning model that underwent supervised training with open-source, labelled textual data on genuine and fake reviews. The tool has been applied to certain ongoing investigations involving digital platforms to analyse a large volume of online reviews and identify suspicious ones.

Again, this is a generic tool which learns to identify text passages with certain characteristics as required by individual use cases. The model can then be trained with corresponding data that are relevant for the model to learn about that required

characteristic. In the future, it is conceivable that the same tool can be applied to detect suspicious tenders involving bid-rigging, or even resumes in the agency's recruitment exercises to identify potential candidates for interviews.

III. AI Verify Toolkit (AIVT)

CCCS is a member of the AI Governance Roundtable, which is a committee between government agencies to devise AI governance principles in Singapore. The AI Verify foundation is a collaboration between technology partners in the public and private sectors to promote best practices and standards for AI. The AI Verify Toolkit is an open-source AI testing tool developed by the AI Verify foundation to test the compliance of AI algorithms with Singapore's AI governance principles.

In collaboration with the Infocomm Media Development Authority of Singapore (IMDA), the Personal Data Protection Commission (PDPC), and the AI Verify Foundation, CCCS is currently developing a customised "vertical" of the AI Verify Toolkit which tests the compliance of an AI algorithm with competition and consumer protection principles in Singapore. It is envisaged that businesses can incorporate the vertical into their compliance programmes, on a voluntary and self-assessment basis.